

Part-A

1.	Programmer A and Programmer B are working on a software project. The number of lines of code they write per day for 6 days is as follows: Programmer A: [50, 60, 45, 55, 70, 65] Programmer B: [45, 55, 50, 60, 65, 70] (i) Calculate the Pearson correlation coefficient between the productivity of Programmer A and Programmer B based on the number of lines of code they write each day and comment on your result. (ii) Draw a scatter diagram to represent the above information.	Correlation
2.	You are analyzing the performance of a computer algorithm. You have collected data on the input size (x) and the corresponding execution time (y) for different input sizes: x: [100, 120, 140, 150, 160] y: [12, 22, 28, 31, 40] (i) Determine the regression line of y on x for this data. (ii) Estimate the execution time for an input size of 180. (iii) Compute the regression line of x on y for this data. (iv) Predict the input size for an execution time of 45	Regression
3.	State Bayes theorem. Three routers — R1, R2, and R3 — forward 40%, 35%, and 25% of the total data packets respectively. The probability of packet loss for R1 is 0.5%, for R2 is 1%, and for R3 is 1.5%. If a packet is found to be lost, find the probability that it was routed through: (i) R2? (ii) R3?	Fundamentals of Probability
4.	A company has three servers — Server A, Server B, and Server C — handling 30%, 45%, and 25% of the total requests respectively. Server A fails 1% of the time, Server B fails 2%, and Server C fails 3%. A request is found to have failed. What is the probability that it was handled by: (i) Server A? (ii) Server C?	Fundamentals of Probability

Part-B

5	A continuous random variable X has the following probability density function: $f(x) = \frac{1}{24} (x^2 + 1); 1 < x < 4$ Compute (i) the value of $E(X)$ (ii) $SD(X)$ (iii) $P(x < 3.5)$ and (iv) $P(x > 0.3)$	Random variable and mathematical expectation
6	A continuous random variable X has the following probability density function: $f(x) = K(x - 1); 2 \leq x \leq 6$ Compute (i) the value of 'K' (ii) $E(x)$ (iii) $CV(x)$ and (iv)	Random variable and mathematical expectation

7.	Suppose that in a certain region of a country the daily rainfall (in inches) is a continuous random variable X with probability density function f(x) given by $f(x) = \frac{3}{12}(6x - 3x^2) \quad , 0 < x < 2$ Find the probability that at a given day in this region the rainfall is (i) not more than 1.5 inches. (ii) between 0.5 and 1.5 inches. Also calculate mean and variance of the daily rainfall (in inches).	Random variable and mathematical expectation									
8.	A software engineer records an average of 5 login failures per hour on the university portal. Find the probability that i) Exactly 4 login failures occur ii) At most 2 login failures occur iii) At least 3 login failure occur.	Probability Distribution									
9.	In a software development company, a project manager is interested in investigating whether there is a correlation between the type of programming language used (Python or Java) and the time of day when coding errors occur (morning or evening). The project manager collects data on coding errors over a month and constructs a 2x2 contingency table as follows: <table border="1"> <tr> <td></td><td>Python</td><td>Java</td></tr> <tr> <td>Morning</td><td>40</td><td>15</td></tr> <tr> <td>Evening</td><td>25</td><td>30</td></tr> </table> Conduct a chi-square test at a significance level of 5% to determine whether there is an association between the programming language used and the time of day when coding errors occur. (Note: At 5% level of significance tabulated value of Chi-square @ 1df = 3.84)		Python	Java	Morning	40	15	Evening	25	30	Estimation and Hypothesis
	Python	Java									
Morning	40	15									
Evening	25	30									